

UNIT II

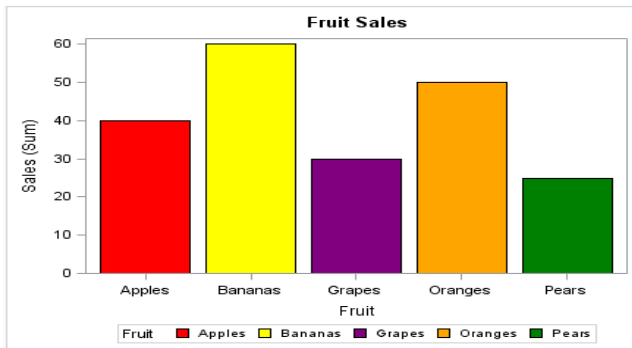
Representation of Data

June 22, 2026

Simple Bar Diagram

Definition

A Simple Bar Diagram is a graphical representation of data using rectangular bars of equal width, where the height (or length) of each bar is proportional to the value it represents.



Steps to Draw a Simple Bar Diagram

- 1 A Simple Bar Diagram consists of vertical or horizontal bars of equal width.
- 2 The length of bars is taken proportionally to the magnitudes of values represented.
- 3 The width of bars has no significance.
- 4 Vertical bars are used to represent quantitative data, whereas horizontal bars are used to represent qualitative data.
- 5 If the data do not relate to time, then they should be arranged in ascending or descending order of magnitude.

Problems on Simple Bar Diagram

1. Draw a simple bar diagram for the following data showing the number of students in different classes.

Class	Number of Students
I	25
II	35
III	30
IV	40
V	50

Problems on Simple Bar Diagram

2. Draw a simple bar diagram representing the production of rice (in tonnes) during five years.

Year	Production
2020	120
2021	150
2022	180
2023	160
2024	200

Problems on Simple Bar Diagram

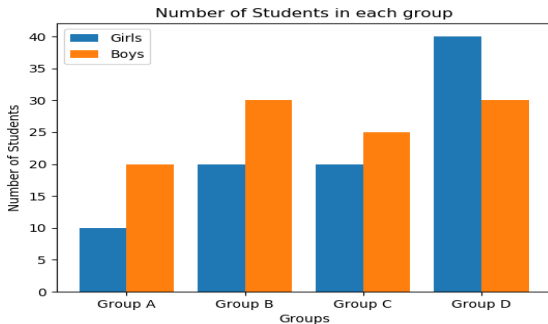
3. Draw a simple bar diagram showing the sales of books in different months.

Month	Books Sold
January	80
February	100
March	120
April	90
May	140

Multiple Bar Diagram

Definition

A Multiple Bar Diagram is a graphical representation in which two or more sets of related data are represented by bars placed side by side for easy comparison.



Steps to Draw a Multiple Bar Diagram

- Multiple Bar Charts are an extension of the Simple Bar Chart.
- Grouped bars are used to represent related sets of data.
- Example: Imports and Exports of a country are shown using a Multiple Bar Chart.
- Each bar in a group is shaded or coloured differently for the sake of distinction.
- Multiple Bar Diagrams are used to denote more than one phenomenon.
- Multiple Bar Charts are useful for direct comparison between two or more values.
- The bars are drawn side by side.
- A key or legend must be given to identify and understand the different bars.

Problems on Multiple Bar Diagram

1. Draw a Multiple Bar Diagram to compare the number of boys and girls in different classes.

Class	Boys	Girls
X	40	35
XI	50	45
XII	60	55

Problems on Multiple Bar Diagram

2. Draw a Multiple Bar Diagram showing the Imports and Exports (in Crores) of a country.

Year	Imports	Exports
2020	120	100
2021	150	130
2022	180	170
2023	200	190

Problems on Multiple Bar Diagram

3. Draw a Multiple Bar Diagram showing the marks obtained by two students in different subjects.

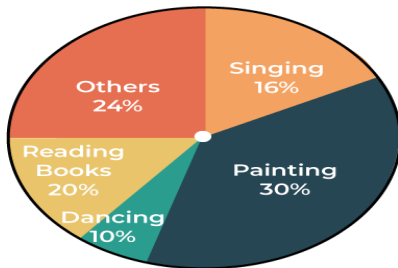
Subject	Student A	Student B
Mathematics	85	80
Physics	78	75
Chemistry	82	88
English	90	85
Computer Science	95	92

Pie Diagram

Definition

A Pie Diagram (Pie Chart) is a circular diagram divided into sectors (slices), where each sector represents a proportion or percentage of the total data.

Number of Students



Steps to Draw a Pie Diagram

- A Pie Chart is used to compare the relation between the whole and its components.
- In a Pie Chart, the area of the sectors of a circle is used to represent the data.
- In a Pie Chart, the circle is drawn with radius proportional to the square root of the quantity to be represented because the area of a circle is πr^2 .
- The sectors are coloured and shaded differently for easy identification.
- To construct a Pie Chart, a circle of suitable radius is drawn and the angles for each sector are calculated as follows:

$$\text{Angle for each sector} = \frac{\text{Component Part}}{\text{Total}} \times 360^\circ$$

- The area of each sector represents the percentage share of the corresponding category.

Problems on Pie Diagram

1. Draw a Pie Chart for the following monthly expenditure of a family.

Item	Expenditure (in Rupees)
Food	4000
Rent	6000
Education	2000
Transport	1000
Savings	2000

Problems on Pie Diagram

2. Construct a Pie Chart showing the marks obtained by a student in different subjects.

Subject	Marks
Mathematics	80
Physics	70
Chemistry	60
English	50
Computer Science	100

Problems on Pie Diagram

3. Draw a Pie Chart representing the population of different villages.

Village	Population
A	3000
B	2500
C	2000
D	1500
E	1000

Definition

A Histogram is a graphical representation of a continuous frequency distribution. It consists of a series of adjacent rectangles where:

- The base of each rectangle represents the class interval
- The height of each rectangle is proportional to the frequency of the corresponding class.
- The area of each rectangle represents the frequency.

Steps to Draw a Histogram

- Mark the class intervals on the X-axis.
- Mark frequencies on the Y-axis.
- Draw rectangles with bases equal to class intervals.
- Make heights proportional to frequencies.
- Rectangles touch each other (no gaps).

Uses of Histogram

- Represents continuous frequency distributions graphically.
- Helps identify the modal class.
- Provides a basis for constructing a Frequency Polygon.
- Helps visualize the shape and spread of data.

Important Remarks

- 1 For an ungrouped frequency distribution, it is assumed that the frequency corresponding to a value is spread uniformly over a suitable interval around that value.
- 2 If the grouped frequency distribution is not continuous, it must first be converted into a continuous distribution before drawing the histogram.
- 3 The height of a rectangle represents the frequency density, while the area represents the actual frequency.
- 4 Histogram is useful in studying the shape of a frequency distribution and identifying the modal class.

Class Boundaries

If d is the gap between the upper limit of one class and the lower limit of the succeeding class, then

$$\text{Upper Class Boundary} = \text{Upper Class Limit} + \frac{d}{2}$$

$$\text{Lower Class Boundary} = \text{Lower Class Limit} - \frac{d}{2}$$

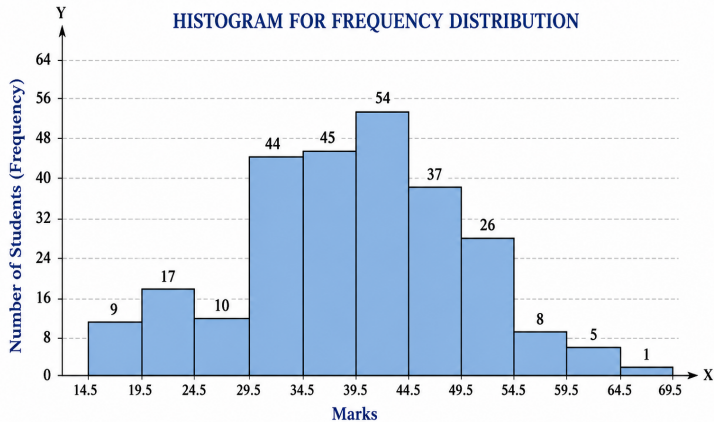
Example

The histogram of the distribution of marks of 250 students is obtained as follows.

Marks	Frequency
15–19	9
20–24	17
25–29	10
30–34	44
35–39	45
40–44	54
45–49	37
50–54	26
55–59	8
60–64	5
65–69	1

Since the grouped frequency distribution is not continuous, we first convert it into a continuous distribution as follows:

Marks	Frequency
14.5–19.5	9
19.5–24.5	17
24.5–29.5	10
29.5–34.5	44
34.5–39.5	45
39.5–44.5	54
44.5–49.5	37
49.5–54.5	26
54.5–59.5	8
59.5–64.5	5
64.5–69.5	1



Marks (Class Interval)	14.5– 19.5	19.5– 24.5	24.5– 29.5	29.5– 34.5	34.5– 39.5	39.5– 44.5	44.5– 49.5	49.5– 54.5	54.5– 59.5	59.5– 64.5	64.5– 69.5
Number of Students (Frequency)	9	17	10	44	45	54	37	26	8	5	1

Frequency Polygon

Definition

A Frequency Polygon is a graphical representation of a frequency distribution. It is especially useful for comparing different data sets and for visualizing the shape of the data distribution more smoothly than a histogram.

Frequency Polygon

A Frequency Polygon is obtained by plotting points where:

- The X-axis (abscissa) represents the mid-values (class marks) of the class intervals.
- The Y-axis (ordinate) represents the frequencies of the corresponding classes.

Steps to Construct a Frequency Polygon

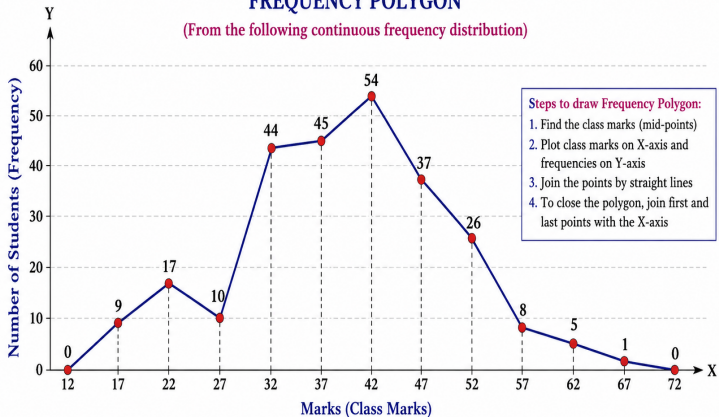
- 1 Calculate class marks:

$$\text{Class Mark} = \frac{\text{Lower Limit} + \text{Upper Limit}}{2}$$

- 2 Plot class marks on the X-axis and frequencies on the Y-axis.
- 3 Join the plotted points with straight lines.
- 4 Add one extra class at each end with zero frequency.
- 5 Join these points to the X-axis to close the polygon.

FREQUENCY POLYGON

(From the following continuous frequency distribution)



Class Interval	14.5–19.5	19.5–24.5	24.5–29.5	29.5–34.5	34.5–39.5	39.5–44.5	44.5–49.5	49.5–54.5	54.5–59.5	59.5–64.5	64.5–69.5
Frequency (f)	9	17	10	44	45	54	37	26	8	5	1
Class Mark (x)	17	22	27	32	37	42	47	52	57	62	67

Frequency Curve

Definition

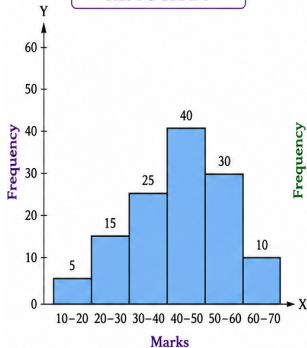
A Frequency Curve is a smooth freehand curve obtained by joining the points of a frequency polygon smoothly. It represents the frequency distribution in a continuous and smooth form.

Steps to Construct a Frequency Curve

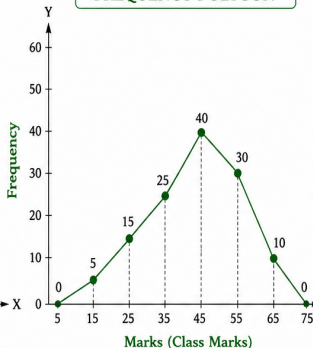
- Find the class marks (midpoints) of the class intervals.
- Plot class marks on the X-axis and frequencies on the Y-axis.
- Plot the corresponding points.
- Draw a smooth curve passing through the points.
- Extend the curve to meet the X-axis at both ends.

HISTOGRAM, FREQUENCY POLYGON AND FREQUENCY CURVE

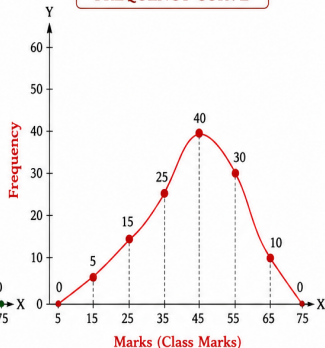
HISTOGRAM



FREQUENCY POLYGON



FREQUENCY CURVE



Class Interval	10-20	20-30	30-40	40-50	50-60	60-70
Frequency (f)	5	15	25	40	30	10
Class Mark (x)	15	25	35	45	55	65

Key Points:

- Histogram is drawn using class intervals on the x-axis and frequencies on the y-axis.
- Frequency polygon is drawn by joining the class marks with their frequencies using straight lines.
- Frequency curve is a smooth curve drawn through the points of the frequency polygon.

Histogram → Frequency Polygon → Frequency Curve

Ogive Curve

Definition

An Ogive is a graph of cumulative frequencies plotted against class boundaries. It is also called a Cumulative Frequency Curve.

Steps to Construct a Ogive Curve

- Calculate cumulative frequencies.
- Draw X-axis and Y-axis.
- Plot class boundaries against cumulative frequencies.
- Join the points by a smooth curve.
- For median determination, draw both less than and more than ogives on the same graph.

Uses of Ogive Curve

- Finding Median.
- Finding Quartiles.
- Finding Percentiles and Deciles.
- Studying cumulative frequencies.
- Determining observations below or above a given value.
- Comparing cumulative distributions.